

DE 424: Practical 10 Report

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Declare that this work presented is your own work. Mention any sources that contributed to work or report
I declare that this report is my own work. This report was written using LaTeX and thus Gemini was used to assist in certain formatting. This includes help to format code listings, mathematical equations and image display. The mathematical derivations and interpretation of results are my own work.

ECSA graduate attributes

Discuss how the GAs were assessed in the module + which parts of the report this is documented. State HOW and WHERE they were satisfied This section may (and should) be brief, but should discuss all aspects of the GAs being assessed. For example, GA1 partially assesses your ability to research literature as part of problem solving, which means that your report must mention and cite (at least) a few sources - can include prescribed resources. Please consult the module framework for a more detailed description of these two GAs as well as how they are assessed in this module.

GA 1: Problem solving

GA 2: Application of scientific and engineering knowledge

1 Problem Description

This project addresses an object tracking task within a discrete grid world inhabited by a latent entity known as a wumpus. The objective is to determine the trajectory of the wumpus across multiple time steps based on inaccurate measurements. At each time step the wumpus attempts to move to an adjacent cell (up, down, left, or right) with equal probability. If the target cell is within the grid boundaries, the wumpus transitions to that location; otherwise it remains in its current position. The exact location of the wumpus is never directly observed. Instead, every cell in the grid generates a binary detection value at each time step, parameterised by:

- True Positive (p_w): The probability that the cell containing the wumpus correctly generates a detection.
- False Positive (p_c): The probability that a cell not containing the wumpus generates an incorrect detection (clutter measurement).

Given the stochastic grid detection data across multiple time steps, the task is to design, implement and evaluate a Probabilistic Graphical Model (PGM) solution. This involves constructing a model that accurately captures the wumpus's transition and observation logic to infer it's latent trajectory across various datasets.

2 Solution Design

The solution was developed iteratively to accomodate the increasing complexity and varying characteristics of each dataset. The PGM design process followed a structured methodology: defining the random variables, constructing the factors, and building a cluster graph for inference. (note: this section must show model structure and show all factors. State and motivate all design choices)

2.1 Dataset 1

The initial model was designed for a 5×5 grid over a series of $T = 10$ time steps. In this scenario, the known parameters $p_w = 0.95$ and $p_c = 0.05$ are constants.

2.2 Random Variable Definition

A critical design choice was made to represent the wumpus's location using a single latent RV W^t instead of separate X^t and Y^t coordinates. The grid is mapped to a 1-dimensional vector space where each cell index $i \in \{0, \dots, 24\}$ corresponds to a unique (x, y) coordinate. This design choice is motivated by the following advantages:

- Reduced complexity: Capturing position in a single RV simplifies factor construction since the complexities of X and Y coordinate dependencies are avoided.
- Enforced physical constraints: Illegal moves (such as diagonal moves) are easily set to a probability of zero within a 25×25 transition matrix.
- Inference efficiency: Less RVs result in smaller clusters in the cluster graph, which leads to more efficient message passing and overall faster convergence.

The RVs are defined as follows:

- W^t : Position of the wumpus as time t , $W^t \in \{0, 1, \dots, 24\}$.
- D_w^t : Detection of wumpus at cell $W^t \in \{0, 1, \dots, 24\}$ at time $t \in \{0, 1, \dots, 9\}$.

2.3 Define Factors

Furthermore, because the resulting structure is a junction tree, the algorithm is guaranteed to reach exact inference in only two passes (forward and backward), ensuring maximum computational efficiency and faster overall runtime.

3 Code Implementation

Discussion + code snippets

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Description of how you processed the output of solution, the presentation of results and analysis thereof. Report both qualitative and quantitative results.